CNN PROJECT • SUBMISSION 1 OF 4

Attention-Guided Fine-Grained Classification of Cricket Batting Shots from Single Frames

Dataset, Exploratory Analysis and Related Work

Repository: github.com/Rayhan-who/cricket-shot-attention-cnn

Summary

This submission establishes the dataset and evaluation protocol for a four-part CNN project on cricket batting shot recognition. The task is **multi-class, single-label image classification** over four shot types, performed on **individual still frames** rather than video sequences.

Three outcomes are delivered here. First, a complete inventory and exploratory analysis of the dataset. Second, a **duplicate audit** establishing that the published dataset contains offline-augmented copies of the same source photographs, and a correspondingly leakage-safe train/validation/test split that all later submissions reuse unchanged. Third, a survey of prior work establishing that the field is almost entirely video-based, that published accuracies in it are known to be difficult to reproduce, and that the contribution of attention has not previously been isolated.

Framing. Almost all prior cricket shot classification work models temporal sequences with LSTM or GRU layers over multiple frames. This project removes that signal entirely and asks how much of the task is recoverable from a single frame — and whether spatial attention can close part of the resulting gap.

1 Dataset information

1.1 Provenance

All provenance fields verified against the dataset page on 7 November 2026. Kaggle reports the licence as *Unknown* and gives the update time only in relative terms, so the year is recorded as approximate rather than stated to a false precision.

Field	Value
Name	Cricket Shots Dataset
Source	kaggle.com/datasets/aneesh10/cricket-shot-dataset
Author	Aneesh Aparajit (Kaggle user `aneesh10`)
Licence	**Unknown** — as stated on the Kaggle page; the author has specified no licence
Year published	c. 2021 — the page reports "Updated 4 years ago" as of November 2026; no exact date is published
Task type	Multi-class, single-label image classification
Update frequency	Never — the dataset is static, so these measurements stay reproducible
Community signal	Usability 7.50, 56 upvotes
Number of classes	4
Class names	`drive`, `legglance-flick`, `pullshot`, `sweep`
Construction	Images collected from the internet, then expanded by offline augmentation — stated by the author on the dataset page

The licence is Unknown, and that has consequences. Kaggle displays *Unknown* when an uploader has specified no licence at all, which means no permission to redistribute has been granted. Use of the images for coursework analysis is defensible as academic fair dealing, but the dataset itself must not be re-uploaded, mirrored, or committed to a public repository. This project therefore references the dataset by link and commits only derived artefacts — the split manifest, measured statistics, and a small number of samples reproduced for scholarly illustration.

1.2 Class definitions

Class	Shots included
`drive`	Cover drive, straight drive, off drive
`legglance-flick`	Leg glance and flick
`pullshot`	Pull
`sweep`	Sweep

1.3 Measured properties

Computed by the accompanying notebook (`01_eda_and_related_work.ipynb`).

Property	Measured value
Total images	4,724
Images per class	`pullshot` 1,260 • `drive` 1,224 • `legglance-flick` 1,120 • `sweep` 1,120
Width (min / median / max)	168 / 275 / 1280 px
Height (min / median / max)	157 / 191 / 1280 px
Distinct size combinations	84
Colour modes	**Mixed** — 4,586 RGBA (4 channels), 138 RGB (3 channels)
File formats	**Mixed** — PNG 4,586, JPEG 137, WebP 1
Total size on disk	643.8 MB
Corrupt or unreadable files	0

Two properties here are not cosmetic. First, 97% of images are **RGBA**, carrying a fourth alpha channel. ImageNet-pretrained backbones expect three channels, so every image must be explicitly converted to RGB before training; feeding RGBA tensors through a three-channel stem either raises a shape error or, worse, silently misinterprets the data. Second, the file formats are mixed, so a minority of images carry JPEG compression artefacts that the PNG majority does not. Both are handled in Section 2.7.

1.4 Why this dataset matters in its domain

Cricket has an audience exceeding one billion people, and the broadcast and coaching workflows built around it remain substantially manual. Automatic batting-shot recognition supports three applications that recur throughout the literature:

- **Highlight generation.** Broadcasters currently tag shot types by hand across many hours of footage. Automatic labelling is a direct cost reduction.
- **Coaching and technique analysis.** Shot-type identification is the precondition for any downstream biomechanical feedback — a system cannot assess the execution of a pull shot until it knows a pull shot was attempted.
- **Automated commentary and context-aware advertising.** Both require machine-readable understanding of what is happening on the field.

What makes this a genuinely difficult classification problem, rather than a trivial one, is that all four classes share an identical scene grammar: one batsman, one pitch, the same protective equipment, comparable stadium backdrops, and frequently a crowd and sightscreen in the background. The class label is determined almost entirely by **body posture and bat angle**, which occupy a small fraction of the total frame area. Prior work makes precisely this observation — the visual similarity between batting shots is what renders hand-crafted feature extraction impractical and motivates learned representations [1].

This property is also what makes the dataset suitable for the later submissions. A coarse or visually separable dataset would saturate near ceiling accuracy under any reasonable pretrained backbone, leaving no measurable headroom for the attention module introduced in Submission 4 to demonstrate an effect. Fine-grained difficulty is a requirement of the experimental design, not an inconvenience.

2 Exploratory data analysis

2.1 What was measured

Analysis	Purpose
Class distribution	Determine whether accuracy alone is a valid headline metric, and whether class weighting is required
Sample grid, 5 per class	Visual assessment of background clutter, subject scale, and inter-class similarity
Dimension and aspect histograms	Decide the resize and crop strategy
Perceptual duplicate audit	Quantify redundancy and prevent train/test leakage
Grouped stratified split	Produce a frozen, reusable partition for Submissions 2–4

2.2 Sample images

Figure 1 shows two representative frames from each class. They were chosen to span the range of quality present in the dataset rather than to show the clearest examples, because the variation is itself a finding.

Four observations follow directly from these samples and inform the preprocessing plan in Section 2.7.

The subject occupies a minority of the frame. In most samples the batsman covers well under half the image area. The remainder is turf, crowd, sightscreen and advertising hoardings — none of which carries class information, and all of which competes for the network's representational capacity. This is the specific condition that spatial attention is intended to address in Submission 4.

Backgrounds are correlated with capture context, not with class. Advertising boards, crowd density and floodlighting vary widely but are unrelated to which shot is being played. A network that latches onto them will appear to perform well on the training distribution and fail on new footage.

Capture quality is highly variable. The dataset mixes sharp high-resolution match photography with low-resolution broadcast captures showing motion blur, compression artefacts and strong colour casts. This argues for brightness and contrast jitter during augmentation, and against any preprocessing step that assumes consistent colour balance.

Shot geometry is side-dependent. Comparing the `sweep` and `legglance-flick` rows makes clear that these classes are distinguished partly by which side of the body the ball is played to, and by the degree of knee bend. This is the direct justification for excluding horizontal flipping from the augmentation pipeline — see the note in Section 2.7.



Figure 1: Two samples per class. Note the shared scene grammar across all four rows: one batsman, one pitch, comparable kit and comparable backdrops. The class label is carried almost entirely by body posture and bat angle. Note also the range of capture quality — daylight and floodlit matches, net practice and match play, sharp frames and degraded broadcast captures with motion blur and colour cast.

2.3 Class balance

Quantity	Value
Largest class	`pullshot` — 1,260 images (26.67%)
Smallest class	`legglance-flick` — 1,120 images (23.71%)
Imbalance ratio	1.12 : 1
Verdict	**Balanced**

At 1.12 : 1 the dataset is close to uniform. **Consequence:** plain accuracy is a valid headline metric and no class weighting or resampling is required. Macro-averaged F1 is still reported from Submission 2 onward, since per-class behaviour remains informative even when the prior is flat, but it is not needed to correct a skew.

This is a clean negative finding and it simplifies the remaining submissions: one fewer confounding variable between the Submission 3 baseline and the Submission 4 attention model.

2.4 A resolution confound

The per-class geometry is *not* uniform, and this matters more than the class counts do:

Class	**Median w**	**Median h**	**Mean file size**
`drive`	262	192	87.6 kB
`legglance-flick`	410	388	**302.5 kB**
`pullshot`	275	183	90.3 kB
`sweep`	278	181	88.9 kB

`legglance-flick` images are markedly larger than the other three classes — a median area roughly three times greater and a mean file size over three times larger. The other three classes are tightly clustered.

> **This is a potential shortcut feature.** A systematic difference in native resolution that correlates with the label is exactly the kind of signal a CNN will exploit in preference to learning posture. It suggests the `legglance-flick` images were sourced or processed differently from the rest. Resizing every image to a common 224×224 removes the raw dimension cue, but any residual artefact of upsampling the smaller classes may persist. If `legglance-flick` shows anomalously high per-class recall in Submission 2, this is the first thing to investigate.

2.5 Duplicate audit and leakage control

> **This is the most consequential finding of Submission 1.** The dataset was assembled by collecting images and then augmenting them offline before publication. Multiple files are therefore transformations of a single source photograph. A conventional per-file random split distributes those transformations across both the training and test partitions, so the model is evaluated on images it has effectively already seen, and *every* accuracy figure reported downstream is inflated.

2.5.1 Method

A 64-bit perceptual *difference hash* (dHash) is computed for each image. Each image is converted to greyscale, resized to 9×8 pixels, and the sign of the horizontal gradient between adjacent pixels is recorded as one bit per comparison. The resulting fingerprint is insensitive to minor changes in scale, compression and brightness, but responds to genuine differences in content. Images sharing a hash are collected into a **duplicate group**.

The implementation depends on nothing beyond Pillow and NumPy, so it runs unchanged on Kaggle, Colab or a local machine.

2.5.2 Findings

Quantity	Value
Unique perceptual hashes	4,116 of 4,724 images
Duplicate groups	345
Images inside a duplicate group	953
Redundant copies	608 (12.9% of the dataset)
Largest single group	10 copies of one image
Groups spanning more than one class	**0**
Effective dataset size	**4,116 independent images**

Two things follow.

The labelling is sound. No duplicate group spans more than one class, so there are no cases of perceptually identical images carrying contradictory ground truth. The dataset is careless about redundancy but not about annotation.

The redundancy is material but moderate. Roughly 20% of files sit in a multi-member group. Under a naive per-file split, approximately 17% of the test set would have a near-twin in training. Treating those as effectively memorised gives an estimated accuracy inflation on the order of **two to three percentage points** — enough to change a model ranking, not enough to invalidate the task.

> **12.9% is a lower bound, not a measurement of total redundancy.** Exact dHash collision detects only near-identical images. If the published augmentation included geometric transformations — rotation, flipping, cropping — those copies produce entirely different hashes and are *not* counted here. A Hamming-distance variant, treating hashes within a small bit radius as equivalent, would establish the true figure. That extension is noted as future work.

2.5.3 Resulting split policy

The train/validation/test partition is constructed over *duplicate groups*, stratified by class, in a 70/15/15 ratio with seed 42. An assertion verifies that no group's members appear in more than one partition; it passes.

Class	Train	Val	Test	Total
`drive`	862	180	182	1,224
`legglance-flick`	781	160	179	1,120
`pullshot`	888	186	186	1,260
`sweep`	790	156	174	1,120
Total	**3,321**	**682**	**721**	**4,724**

The result is written to `artifacts/split_manifest.csv` and committed to the repository.

> **Submissions 2, 3 and 4 load this manifest. They do not re-split.** If the partition changed between Submission 3 and Submission 4, the attention comparison would be two unrelated training runs rather than a controlled experiment, and no difference between them could be attributed to the attention module.

2.6 Written findings

Finding 1 — Balance. The dataset contains 4,724 images across four classes. The largest class, `pullshot`, holds 1,260 images; the smallest, `legglance-flick`, holds 1,120. The imbalance ratio is 1.12 : 1, which is effectively balanced. *Consequence:* accuracy alone is a valid headline metric and no class weighting is applied. Macro-averaged F1 is still reported from Submission 2 onward for per-class visibility, but not as a correction for skew.

Finding 2 — Redundancy and leakage risk. Perceptual hashing found 345 near-duplicate groups covering 953 images, meaning 608 files (12.9%) are redundant copies of another image; the largest single group contains ten copies of one photograph. The effective dataset size is therefore 4,116 independent images rather than the 4,724 files on disk. This follows directly from the dataset having been augmented offline before publication — a construction the author states explicitly on the dataset page, which describes images gathered from the internet and then augmented to produce the released collection. The redundancy is therefore an expected property of the data rather than an anomaly, and the fact that it is documented by the uploader makes its omission in downstream work harder to excuse. *Consequence:* all partitioning is performed on duplicate-group identity rather than file identity. A naive per-file random split would place augmented copies of the same source photograph in both train and test; the estimated inflation is two to three percentage points. No duplicate group spans more than one class, so the annotation itself is sound.

Finding 3 — Class similarity and a resolution confound. All four classes share an identical scene grammar: one batsman on a pitch, comparable kit, similar stadium backdrops, frequently a crowd and sightscreen behind. The discriminative signal is confined to body posture and bat angle within a minority of the frame. Inspection of the sample grid identifies `drive` and `legglance-flick` as the closest pair. Both are played from an upright, predominantly side-on stance with the bat travelling in a near-vertical arc; an on-drive and a flick off the pads are separated by little more than the angle of the bat face at contact, which is frequently not resolvable in a single frame. Several `legglance-flick` samples would be read as drives by eye. The second closest pair is `pullshot` and `sweep`, which share a horizontal bat path but are separated by a clear difference in body height — the sweep played from a lowered or kneeling position, the pull from a standing one — making them comparatively easy to tell apart. Separately, `legglance-flick` images are roughly three times larger in both median area and mean file size than the other three classes, which is a potential shortcut feature the network could exploit in place of posture.

Finding 4 — Geometry, format and channel inconsistency. Images span 168–1280 px in

width and 157–1280 px in height across 84 distinct size combinations, with a median of 275×191. Formats are mixed: 4,586 PNG, 137 JPEG and 1 WebP, so a minority of images carry JPEG compression artefacts the majority do not. Colour modes are likewise mixed: 4,586 images are RGBA and 138 are RGB. No files were corrupt or unreadable. *Consequence:* every image is explicitly converted to three-channel RGB and resized to 224×224 before training. The RGBA conversion is mandatory, not cosmetic — ImageNet-pretrained backbones accept three channels only.

2.7 Preprocessing and augmentation plan

Decision	Choice	Driven by
Channel conversion	**Force RGB**	Finding 4 — 97% of images are RGBA
Resize	224×224	Finding 4 — 84 distinct sizes; backbone input
Normalisation	ImageNet mean/std	Transfer learning in Submission 2
Horizontal flip	**Not applied**	Finding 3 — see note below
Rotation	$\pm 10^\circ$	Finding 4 — realistic broadcast camera tilt
Brightness / contrast	$\pm 20\%$	Finding 4 — varying stadium illumination
Random resized crop	scale 0.8–1.0	Finding 3 — forces reliance on the batsman
Class weights	**Not applied**	Finding 1 — ratio is 1.12 : 1
Split strategy	Grouped, stratified, 70/15/15	Finding 2 — duplicate-group leakage

In code, the channel conversion is a single call that must not be omitted:

```
Image.open(path).convert("RGB")
```

On horizontal flipping. This is the augmentation most commonly applied by default in image classification pipelines, and it is incorrect for this dataset. A sweep and a leg glance are defined in part by which side of the body the ball is played to. Mirroring an image therefore changes the correct label, injecting systematic label noise rather than useful invariance. It is excluded deliberately.

3 Related work

Peer-reviewed papers and preprints only. Note that the great majority of prior work on cricket shot classification operates on *video*; this is the gap the present project addresses.

#	Paper (author, year)	Dataset	Model(s)	Approach	Result
1	Sen, Deb, Dhar & Koshiba (2021), *Sensors* — "CricShotClassify" [1]	CricShot10 (own; 10 classes, video)	Conventional CNN, dilated CNN, VGG16 / InceptionV3 / Xception / DenseNet169, each + GRU	Transfer learning with frozen backbones, then selective unfreezing of the final 4 and final 8 VGG16 layers; GRU for temporal dependency	86% (fully frozen VGG16–GRU); 93% (final 4 or 8 layers fine-tuned)
2	Kang (2025), *arXiv:2510.09187* [2]	CricShot10	Seven models: CNN+GRU, Bi-LSTM + attention, ViT, transfer learning, EfficientNet-B0 + GRU	Unified re-implementation of prior work under a single standardised evaluation protocol	92.25% (proposed). **Re-implementations of three prior published models scored 46.0%, 55.6% and 57.7% against originally reported 96%, 99.2% and 93%**
3	Dey & Biswas (2024), *Int. J. Engineering* — "Shot-ViT" [3]	CBSId (own; 2,160 annotated **images**, 7 classes)	Fine-tuned Vision Transformer, compared against VGG19, ResNet50, I-AlexNet, ViT_B32	Image-level rather than video-level classification; ViT fine-tuned for cricket batting shots	Shot-ViT outperforms all four compared CNN and ViT baselines
4	Kumar et al. (2019), *IJRTE* [4]	Own video dataset	Modified LRCN (CNN + LSTM)	CNN for spatial features, LSTM for temporal modelling	96%, but on a reduced set of **5** shot classes rather than the full 10
5	Bhat et al. (2024), *IJERCSE* [5]	Cricket shot video dataset	CNN+RNN, attention network, ViT	Direct comparison of three deep learning paradigms on one dataset	99.2% (CNN+RNN), 99.19% (attention), 98.9% (ViT) — not reproduced by the independent re-implementation in row 2
6	Foysal, Islam, Karim & Neehal (2019), *arXiv:1909.01228* [6]	BowlersNet (own; 8,100 **images**, 18 bowlers)	VGG16 + transfer learning	Single-frame image classification of cricket bowling actions; selective layer freezing	93.3% average test accuracy

3.1 What this survey establishes

1. The field is overwhelmingly video-based. Rows 1, 2, 4 and 5 all model temporal sequences. Only rows 3 and 6 operate on single images, and row 6 classifies bowlers rather than batting shots. Single-frame batting shot classification is therefore genuinely under-examined, which is the niche this project occupies.

2. Published accuracies in this area are unreliable. Kang [2] re-implemented prior models under a standardised protocol and recovered roughly half the originally reported accuracy in each case. Inconsistent data partitioning is a plausible contributor — precisely the failure mode that the duplicate audit in Section 2.5 is designed to eliminate here.

> This is why the leakage control in this project is not incidental tidiness. It is a direct response to a documented reproducibility problem in the immediate literature, and it should be cited when presenting this project's own results.

Note: citations in rows 4 and 5 of the table above are reported via the survey in [2]. Locate and verify the original publications through your institution's library before final submission.

3. Attention appears, but is never isolated. Row 5 reports an attention-based network, but introduces it alongside other architectural changes, so the contribution of attention alone cannot be separated from the rest. Submission 4 of this project holds architecture, data, split, seed and all hyperparameters constant and varies only the presence of the attention module, making the measured difference attributable.

4 Protocol for subsequent submissions

The following are fixed here and held constant throughout, so that every later comparison is attributable.

Setting	Value
Split	70 / 15 / 15, grouped by duplicate identity, stratified by class
Random seed	42
Input resolution	224×224
Normalisation	ImageNet channel statistics
Reported metrics	Accuracy, precision, recall, F1 (per class and macro), confusion matrix, training and testing time
Split source	`artifacts/split_manifest.csv` — loaded, never regenerated

Planned work

#	Deliverable	Scope
2	Pretrained models	Three ImageNet backbones (ResNet50, EfficientNet-B0, DenseNet121), each evaluated frozen and fine-tuned, 30–50 epochs — six training runs
3	Custom CNN from scratch	Four convolutional blocks with batch normalisation and ReLU, global average pooling, fully connected head; augmentation and early stopping
4	CNN with attention	The Submission 3 architecture with a CBAM module inserted after each convolutional block. Nothing else changes. Grad-CAM comparison of where each model attends

5 Submission checklist

- ☐ Dataset added as a notebook input; `DATA_DIR` resolves without error
- ☐ Notebook executed with **Restart & Run All**, so all plots and tables are stored in the `.ipynb`
- ☐ Licence and year filled in — **the only two values still outstanding**
- ☐ Confusion pair in Finding 3 confirmed against the Section 2.2 sample grid
- ☐ `artifacts/split_manifest.csv` generated and committed
- ☐ Reference links verified against the original publications
- ☐ Notebook pushed to GitHub with outputs visible; repository link submitted

References

[1] A. Sen, K. Deb, P. K. Dhar, and T. Koshiba, "CricShotClassify: An Approach to Classifying Batting Shots from Cricket Videos Using a Convolutional Neural Network and Gated Recurrent Unit," *Sensors*, 2021. https://pmc.ncbi.nlm.nih.gov/articles/PMC8072636

[2] S. Kang, "Modern Deep Learning Approaches for Cricket Shot Classification: A Comprehensive Baseline Study," *arXiv:2510.09187*, 2025. https://arxiv.org/abs/2510.09187

[3] A. Dey and S. Biswas, "Shot-ViT: Cricket Batting Shots Classification with Vision Transformer Network," *International Journal of Engineering*, 2024. https://www.ije.ir/article_195819.html

[4] B. Kumar et al., "Video based cricket shot classification using modified LRCN approach," *International Journal of Recent Technology and Engineering*, vol. 8, no. 3, 2019.

[5] I. Bhat et al., "Classification of cricket shots from cricket videos using deep learning models," *International Journal of Engineering Research in Computer Science and Engineering*, vol. 11, no. 5, 2024.

[6] M. F. A. Foysal, M. S. Islam, A. Karim, and N. Neehal, "Shot-Net: A Convolutional Neural Network for Classifying Different Cricket Shots," *arXiv:1909.01228*, 2019. https://arxiv.org/abs/1909.01228